CUNY Queens College

Department of Computer Science

CSCI 212 Lab – Object Oriented Programming in Java
Project 1

Objective

In this project, you will build a Java application to manage and analyze student grades for a specific course. You will practice working with:

- Classes and Objects
- ArrayLists
- Arrays
- Static Utility Methods

This will also be strong practice in **MODULARIZATION**, meaning separating code into multiple methods and classes based on their function.

Project Structure

You will create **FOUR classes** that work together as specified below. Make sure to follow this structure carefully.

- **Main.java** – Driver class
- **Student.java** – Blueprint for student data
- **ClassRoomTools.java** – Static utility methods
- **ClassRoom.java** – Manages the list of students

Recommendation: Work step-by-step to avoid becoming overwhelmed.

Class 1: `Student.java`

This class serves as a blueprint for a student's raw data.

Instance Variables

- `name` : Student's full name (`String`)
- `id` : Student's identification number (`int`)
- `homeworkGrades` : Array storing 4 homework grades (`double[]`)
- `examGrades` : Array storing 3 exam grades (Midterm 1, Midterm 2, Final) (`double[]`)

Constructor Requirements

- Accept arguments for all four variables.
- Check that inputs are not `null`.
- Validate grades are between **0 and 100**.

Methods

Create public getters for all fields:

- `getName()`
- `getId()`
- `getHomeworkGrades()`
- `getExamGrades()`

Class 2: `ClassRoomTools.java`

This is a **utility class**. It does not store data. All methods must be **STATIC**.

Required Methods

1. computeStudentHWavg

`public static double computeStudentHWavg(Student student)`

- Compute the average of the student's homework grades.
- Return the result as a double.

2. computeStudentFinalGrade

`public static double computeStudentFinalGrade(Student student)`

Compute the weighted average:

- Midterm 1: 25% (Index 0)
- Midterm 2: 25% (Index 1)
- Final Exam: 35% (Index 2)
- Homework Average: 15%

Example calculation:

$$(90 \times 0.25) + (80 \times 0.25) + (85 \times 0.35)$$

Then add 15% of the homework average.

3. letterGrade

```
public static String letterGrade(double grade)
```

Range	Letter
90–100	A
80–89	B
70–79	C
60–69	D
Below 60	F

4. computeAverageFinalGrade

```
public static double computeAverageFinalGrade(ClassRoom c)
```

- Compute the average final grade of all students.
- Return the result as a double.

Class 3: `ClassRoom.java`

Manages the list of students.

Instance Variables

- `className` : Name of the course (`String`)
- `studentList` : `ArrayList<Student>`

Constructor

- Initialize `className`
- Instantiate empty `ArrayList<Student>`

Methods

addStudent

```
public void addStudent(Student student)
```

- Adds a Student object to the list.

printSummary

```
public void printSummary()
```

- Print classroom average final grade.
- Use `ClassRoomTools.computeAverageFinalGrade(this)`

printAllStudentInfo

`public void printAllStudentInfo()`

For each student, print:

- Name
- ID
- Homework Average
- Numerical Final Grade
- Letter Grade

Class 4: `Main.java`

Driver class to test your program.

Requirements

- Create at least 4 Student objects.
- Create a ClassRoom object (example: `"CS101"`).
- Add students to the classroom.
- Call:
  - `printSummary()`
  - `printAllStudentInfo()`

Submission Requirements

Submit the following on Brightspace:

- All four `.java` files
- PDF or screenshot of your program output

Deadline: Sunday March 8th at 11:59PM

There will be **no extensions** unless there is a documented excuse (e.g., medical).

Important Notes:

- Start early.
- Ask questions if needed.
- You may consult AI for specific questions.
- Do NOT have AI generate your entire project.

CSCI212 Lab : Project #1 — CUNY Queens College, Department of Computer Science

This project is designed to combine everything you have learned so far. You should be writing the code yourself.

Good luck — and build it step by step!